

notes: logical atomism in HOL

September 30, 2026

In this note I want to explore some object language formulations of logical atomism. I will be setting aside the concept of complete independent generation for now—formalized as Induction and Recursion below—and instead focus on a different idea that is a weakening of strong internal freeness. I will be working within the theory $C.\Box RC$, but I will be reasoning informally in English. I will additionally assume the a non-logical constant $E : t \rightarrow t$, meaning ‘elementary’, and will assume WLOG, that E is rigid. Recall that Rigidity proves that every property of propositions has a GLB.

Let $\sigma^n \rightarrow \tau$ stand for $\underbrace{\sigma \rightarrow \dots \rightarrow \sigma}_n \rightarrow \tau$. Define the lower bound and greatest lower bound of a relation $S : t^n \rightarrow t$ as follows and

$$LB = \lambda S p_1 \dots p_n. \forall q_1 \dots q_n (S q_1 \dots q_n \rightarrow \bigwedge_{1 \leq i \leq n} p_i \leq q_i)$$

$$GLB = \lambda S p_1 \dots p_n. LB S p_1 \dots p_n \wedge \forall q_1 \dots q_n (LB S q_1 \dots q_n \rightarrow \bigwedge_{1 \leq i \leq n} q_i \leq p_i)$$

1 Complete Generation

Say that a relation between propositions is closed when it is closed under the boolean operations.

$$\begin{aligned} \text{Closed} := & \lambda R. (\forall p_1 \dots p_n (R p_1 \dots p_n \rightarrow R \neg p_1 \dots \neg p_n) \wedge \\ & \forall_{t^n \rightarrow t} S (S \subseteq R \rightarrow \\ & \forall_t p_1 \dots p_n (GLB S p_1 \dots p_n \rightarrow \\ & R p_1 \dots p_n))) \end{aligned}$$

Induction Any property of propositions containing the elementary propositions and closed under the Boolean operations, contains all propositions.

$$\forall_{t \rightarrow t} X ((E \subseteq X \wedge \text{Closed } X \rightarrow \forall p. X p)$$

Recursion Any two functional relations that agree on the elementary propositions and preserve the Boolean operations, are coextensive.

$$\forall_{t \rightarrow t \rightarrow t} F \forall_{t \rightarrow t \rightarrow t} G (\text{Fun}_{E \lambda p. \top} F \wedge \text{Fun}_{E \lambda p. \top} G \wedge \text{Closed } F \wedge \text{Closed } G \rightarrow F \sim_E G \rightarrow F \sim G)$$

Where $F \sim_E G$ means $\forall p (E p \rightarrow \forall q (F p q \leftrightarrow G p q))$ and $F \sim G$ means $F \sim_{\lambda p. \top} G$.

2 The Tractarian Truth-Table Postulate

Here is the intuition, loosely inspired by the method of truth-tables from the Tractatus:

The Tractarian Truth-Table Postulate

- (a) The truth values of all propositions are uniquely determined by the assignment of truth values to the elementary propositions.
- (b) Any assignment of truth-values to the elementary propositions can be extended to consistent assignment of truth-values to all propositions.

Informally a consistent assignment of truth values is one that assigns truth to a conjunction only if it assigns truth to the conjuncts, and assigns to p the opposite of the truth value it assigns to its negation. In algebraic terms, we are thus looking a weakening of strong internal freeness. Identifying the truth-value space with the algebra $2 = \{\top_B, \perp_B\}$, then the weaker condition is that any function $f : X \rightarrow 2$ extends to a unique complete homomorphism $\hat{f} : B \rightarrow 2$.¹

In the following the domain, range and notion of functionality for binary relations is defined in the usual manner.

Definition 1 (Truth assignment). *A truth-value assignment is a rigid binary relation $V_0 : t \rightarrow t \rightarrow t$ that is (i) functional, (ii) it's domain are all and only the elementary propositions, (iii) only $\top$ and $\perp$ are in its range.*

Definition 2 (Valuation). *A valuation is a rigid binary relation $V : t \rightarrow t \rightarrow t$ that is (i) functional, (ii) it's domain includes every proposition, (iii) only $\top$ and $\perp$ are in its range and (iv) it preserves GLBs and negation, in the sense that for any $X : t \rightarrow t$, $Vp\top$ for all p such that Xp , iff $V(\text{GLB}(X))\top$ where $\text{GLB}(X)$ is the GLB of the Xs , and Vpq iff $V\neg p\neg q$.*

We say a valuation V extends a truth-value assignment V_0 iff for every elementary proposition p , and every proposition q , V_0pq iff Vpq .

Useful lemma:

Lemma 1. *In Classicism + $\Box RC$, you can prove $\exists V(\text{Val } V \wedge \forall p(V[p] = \top \leftrightarrow p))$.*

Proof. Using RC we can find a rigid relation that relates truths to $\top$ and falsehoods to $\perp$. This is a valuation: in particular because we can prove $\Box \text{Actuality}$ from $\Box \text{Rigid Comprehension}$, the GLB of all truths is true, so this functional relation preserves GLBs. $\square$

To preserve intuitions with reasoning about valuations we shall sometimes write $V[p] = q$ to mean Vpq . I will also use the notation $\{p \mid \phi\}$ for the rigid $t \rightarrow t$ property coextensive with $\lambda p.\phi$.

The Truth-Table Postulate Necessarily, every truth-value assignment V_0 extends to a unique valuation V .²

¹As far as I can see, this is neither weaker nor stronger than the original weaker notion of internal freeness. If $f : X \rightarrow B$ is $\top, \perp$ valued, I cannot see that we have a guarantee that $\hat{f} : B \rightarrow B$ is $\top, \perp$ valued.

²The necessarily in front of this postulate is needed for some of the results. But I need to think more about why this is motivated — i guess it follows from the idea that the statements of the Tractatus are meaningless/tautologous.

The existence part of this roughly tells us that every way of assigning truth values to the elementary propositions extends to a consistent way of assigning truth values to all propositions. The uniqueness tells us that the truth-values of all propositions are uniquely determined by the truth values of the elementary propositions.

Remark 1 (Alethic Supervenience). *The is distinct modal way of cashing out the idea that all truths are uniquely determined by the truth values of the elementary propositions. Namely, that the truths of all propositions supervene on the elementary propositions: any complete conjunction of elementary propositions or their negations is (i) consistent and (ii) is such that for any proposition it either entails it or its negation. I think this alethic supervenience can be shown to be equivalent to the truth-table postulate – I want to come back to this.*

Remark 2. *The usual equivalence between sets and characteristic functions can be proven in higher-order logic (provided functions are identified with functional relations). So we can equivalently think of truth-value assignments as sets (rigid properties) of elementary propositions, and valuations as complete consistent sets of propositions.*

Definition 3 (The world according to V). *Let V be a valuation: V tells us, in a consistent manner, what the truth value of every proposition is. The world according to V , or W_V , is the claim that the propositions that V says are true are true, and the propositions V says are false are false. $W_V := \forall p(Vp \top \rightarrow p) \wedge \forall p(Vp \perp \rightarrow \neg p)$. Since V is rigid, this proposition is also the GLB of the propositions p that V says are true with the propositions $\neg p$ that V says are false.*

When V is a valuation, you'd expect W_V to be a world proposition independently of logical atomism. And indeed it is provable in $\text{Classicism} + \Box\text{Rigidity}$ without any further Tractarian postulates.

Proposition 2. *For any valuation V , the world according to V , W_V , is a world proposition (i.e. an atom).*

Proof. W_V is consistent. The reason is that applying V to W_V yields $\top$, yet $V[\perp] \neq \top$, so $W_V \neq \perp$. (The first claim is a matter of turning the following reasoning in to legitimate HOL $V[W_V] = V[\text{GLB}\{p \mid V[p] \top\} \wedge \text{GLB}\{\neg p \mid V[p] = \perp\}] = \text{GLB}\{V[p] \mid V[p] = \top\} \wedge \text{GLB}\{\neg V[p] \mid V[p] = \perp\} = \text{GLB}\top \wedge \text{GLB}\perp = \top$.)

Moreover, for any proposition p , either $V[p] = \top$ or $V[p] = \perp$, and so p or $\neg p$ is a conjunct of W_V , and thus for any proposition p either $\Box(W_V \rightarrow p)$ or $\Box(W_V \rightarrow \neg p)$. (This argument uses the fact that W_V is GLB of the propositions $\{p \mid V[p] = \top\} \cup \{\neg p \mid V[p] = \perp\}$, or equivalently, the fact that V is rigid.)

□

Remark 3. *Indeed, every world proposition W determines a valuation $V[p] = \top$ if $\Box(W \rightarrow p)$ and $V[p] = \perp$ if $\Box(W \rightarrow \neg p)$. So we have more generally:*

World propositions are in one-to-one correspondence with valuations.

Of course, given a mere truth-value assignment, V_0 , we can defined the world according to V_0 analogously, W_{V_0} . While anyone should believe that W_V is a world proposition when V is a valuation, a logical atomist specifically ought to think that W_{V_0} is also a world proposition when V_0 is merely an assignment of truth-values to elementary propositions. The Tractarian

vision is that the truth-values of the elementary propositions alone should be enough to settle the truth-values of all propositions. Thus we should expect W_{V_0} to be consistent and entail any given proposition or its negation. I believe this is a consequence of the Tractarian Truth-Table Postulate. Later I call the claim that W_{V_0} is a world proposition Alethic Supervenience, so the following conjecture would show that the Truth-Table Postulate entails Alethic Supervenience.

Conjecture 3. *The Tractarian Truth-Table Postulate entails that W_{V_0} is a world proposition whenever V_0 is a truth value assignment.*

I can see how to prove this conjecture if we additionally assume BF (but there are still some gaps to fill). I can also prove this without BF provided you strengthen the definition of valuation (and thus strengthen the Truth-Table Postulate) so that it necessarily satisfies the conditions it presently is required to satisfy (the important thing, for this proof, is that it must necessarily have all propositions in its domain, and must be necessarily functional). Moreover, this seems like a natural strengthening to require, and a similar strengthening should be made to the notion of a truth-value assignment.

Proof. This is the proof strategy. Let V_0 be a truth-value assignment, and let V be its unique extension.

First we prove, as above, that W_{V_0} is consistent. As before $V[W_{V_0}] = \top$ and $V[\perp] = \perp$.

We will show that $\Box(W_{V_0} \rightarrow W_V)$. Since the converse entailment is trivial, $W_{V_0} = W_V$, and is thus an atom.

Suppose that W_{V_0} is true. Using $\Box$ Rigidity we may define a functional rigid relation U that maps all the truths to $\top$ and all the falsehoods to $\perp$ — i.e. as the rigid property coextensive with $\lambda pq.(p \wedge (q = \top) \vee (\neg p \wedge q = \perp))$. CLAIM: U is a valuation, since a GLB of truths is true (recall Actuality is a consequence of Rigidity). Also U extends V_0 . CLAIM??: V is still a valuation at the present W_{V_0} possibility (since we don't have BF, we do not know that its domain is all the propositions, so maybe this step fails). Since the Truth-Table Postulate is necessary, and it states the uniqueness of any extension of V_0 , we can infer that $V = U$, and since W_U is true, it follows that W_V is true. Necessitating this reasoning should get us that $\Box(W_{V_0} \rightarrow W_V)$. □

Remark 4. *The above argument makes use of argumentation that is distinctive to the higher-order perspective, and so I do not think it can be turned into an algebraic argument. (Question: can we find a B and $X \subseteq B$ such that every $f : X \rightarrow 2$ extends to a unique complete homomorphism $\hat{f} : B \rightarrow 2$, but such that the proposition $W_{\hat{f}} = \bigwedge_{\hat{f}p=1} p \wedge \bigwedge_{\hat{f}p=0} \neg p$ is not a world proposition?)*

3 Alethic Supervenience

Here is another way of cashing out the idea that the truth-values of all propositions are uniquely determined by the assignment of truth-values to the elementary propositions, and that any such assignment is consistent. It is a modal, as opposed to truth-tabular, condition

Alethic Supervenience Necessarily, for every truth-value assignment V_0 , W_{V_0} is a world proposition:

1. The assignment of truth values according to V_0 is consistent: $\Diamond W_{V_0}$

2. The assignment of truth values according to V_0 settles the truth value of every other proposition: $\forall p(\Box(W_{V_0} \rightarrow p) \vee \Box(W_{V_0} \rightarrow \neg p))$

Proposition 4. *Alethic Supervenience entails that every truth-assignment extends to at least one valuation.*

Proof. Idea for proof. Suppose that V_0 is a truth-assignment. We can define a valuation V by letting $Vp \top$ iff $\Box(W_{V_0} \rightarrow p)$ and $Vp \perp$ iff $\Box(W_{V_0} \rightarrow \neg p)$. We need to check it's a valuation. The negation condition is trivial. To check the GLB preservation condition, let X be some collection of propositions such that $V[p] = \top$ for each p such that Xp . For any given p such that Xp , $\Box(W_{V_0} \rightarrow p)$ (the only other option is that $\Box(W_{V_0} \rightarrow \neg p)$, give the atomicity of W_{V_0} , but this cannot happen since that means $W_{V_0} \wedge p = \perp$, but $V[W_{V_0} \wedge p] = V[W_{V_0}] \wedge V[p] = \top \wedge \top = \top$). So W_{V_0} entails each p such that Xp , and so is a lower bound of the X s. Thus W_{V_0} entails the greatest lower bound of the X s. $\square$

Proposition 5. *Alethic Supervenience entails that every truth-value assignment extends to at most one valuation.*

Proof. Idea.

Suppose that V is a valuation that extends a truth-assignment V_0 .

Claim: for any p if $\Box(W_{V_0} \rightarrow p)$ then $V[p] = \top$, and if $\Box(W_{V_0} \rightarrow \neg p)$ then $V[p] = \perp$. Start with the first case. Suppose $\Box(W_{V_0} \rightarrow p)$, so $W_{V_0} \wedge \neg p = \perp$. On the other hand, suppose for contradiction that $V[p] \neq \top$. So $V[p] = \perp$, and so $V[\neg p] = \top$. So we may compute $V[W_{V_0} \wedge \neg p] = V[W_{V_0}] \wedge V[\neg p] = \top \wedge \top = \top$, yet clearly $V[\perp] = \perp$ so $W_{V_0} \wedge \neg p \neq \perp$, contradiction. The second case is proved in the same way.

Given this claim, then the fact that either $\Box(W_{V_0} \rightarrow p)$ or $\Box(W_{V_0} \rightarrow \neg p)$ for every p means V is uniquely determined by V_0 .

Finally V is a valuation: for any world W , the property of being entailed by W commutes with GLBs and negation, and since this is how V is defined, V has the required properties. $\square$

If all of this checks out, we get the following theorem:

Theorem 6. *Alethic Supervenience entails the Tractarian Truth-Table Postulate.*

And (assuming $\Box BF$, but possibly without) The Tractarian Truth-Table Postulate entails Alethic Supervenience.

4 Atomicity

Theorem 7.

$\Box TTP$ and/or $\Box$ alethic supervenience are consistent with the negation of Atomicity.

This can be established using modalized domains over possibility frames, outlined in my note, *The consistency of Mathematical Possibilism*.

Indeed, in these are models of $C.\Box RC.\Box BF$, so we get the following thing of independent interest.³

³This is shown in my note, *The consistency of Mathematical Possibilism*

Corollary 8. $\Box RC + \Box BF$ and not *Atomicity* are consistent.

Note that give $\Box RC$ you get $\Box$ Actuality, $\Box BF$, and $\Box$ Boolean Completeness. So the theorem is a little surprising.